

MSDS 535 Week 8

Chapter 8 - Cluster analysis: basic concepts and methods

Overview and Objectives

- Understand Cluster analysis basic concepts and methods.
- Partitioning methods.
- Hierarchical methods.
- Density-based and grid-based methods.
- Evaluation of clustering.
- Summarize cluster analysis: basic concepts and methods.

What is cluster analysis?

- Cluster analysis is the process of dividing data objects into subsets (clusters) based on their similarity.
- Different clustering methods can yield different clusterings on the same data set.
- Applications of cluster analysis include business intelligence, image recognition, web search, biology, and security.
- It can be used to group customers with similar characteristics, categorize projects for better management, and cluster photos based on common features.
- Cluster analysis is also used in web search to organize search results and document topics.
- It can provide insights into data distribution, serve as a preprocessing step for other algorithms, and is useful for outlier detection in applications like fraud detection.

Requirements for cluster analysis

- Ability to handle various types of data objects.
- Scalability for large datasets.
- Capability to discover clusters with arbitrary shapes.
- Minimized reliance on user-specified input parameters.
- Robustness to noisy data.
- Support for incremental clustering and insensitivity to input order.
- Handling high-dimensionality data.
- Accommodation of constraint-based clustering.
- Interpretability and usability for practical applications.

Comparison Aspects for Clustering Methods

- Single vs. multilevel clustering.
- Separation of clusters (mutually exclusive or not).
- Choice of similarity measures.
- Clustering in the full data space vs. subspaces.

Overview of basic clustering methods

- **Clustering Categorization:** Clustering methods in data mining are diverse and may overlap in categories. They can be broadly categorized into three groups: partitioning methods, hierarchical methods, and density-based/grid-based methods.
- **Partitioning Methods:** These methods divide a dataset into k partitions (clusters), typically employing distance-based techniques. Popular heuristic algorithms like k -means and k -medoids are commonly used for partitioning. They work well for finding spherical-shaped clusters in small-to-medium-sized datasets.

Overview of basic clustering methods

- **Hierarchical Methods:** Hierarchical clustering creates a hierarchical decomposition of data objects. Agglomerative methods start with individual objects and merge them into clusters, while divisive methods begin with all objects in one cluster and split them into smaller groups. Hierarchical clustering can be distance-, density-, or continuity-based and is rigid in that it cannot correct erroneous decisions.
- **Density-Based and Grid-Based Methods:** These methods consider the density of data points to form clusters. Density-based methods continue growing clusters as long as the density exceeds a threshold, making them suitable for arbitrary-shaped clusters. Grid-based methods quantize the object space into cells, allowing for efficient processing.

Overview of basic clustering methods

- Integration and Overlapping Methods: Some clustering algorithms combine ideas from multiple clustering methods. Furthermore, certain applications may require the integration of various clustering techniques based on specific criteria.

Partitioning methods

- Partitioning is the simplest and most fundamental version of cluster analysis, where objects are organized into exclusive clusters, each with a representative.
- Two key technical challenges in partitioning methods include determining cluster representatives and measuring the distance or similarity between objects and representatives.
- The objective is to create clusters optimized according to a dissimilarity function based on distance, ensuring that objects within a cluster are similar to each other and dissimilar to objects in other clusters.
- The number of expected clusters is the initial parameter for partitioning methods, providing a concise problem specification.
- The section covers partitioning methods, starting with the most prominent k-means approach. It also explores variations for different data types and applications and introduces advanced techniques like kernel k-means for handling nonlinear separability in high-dimensional data.

k-Means: a centroid-based technique

- Partitioning methods distribute objects in a data set into k exclusive clusters, aiming for high intracluster similarity and low intercluster similarity.
- Centroid-based partitioning uses the centroid of a cluster as its representative, defined by the mean or medoid of the objects within the cluster.
- The within-cluster variation measures the sum of squared error between objects and their cluster's centroid, optimizing cluster compactness and separation.

k-Means: a centroid-based technique

- Minimizing within-cluster variation is computationally challenging and often tackled with greedy approaches, like the k-means algorithm, which iteratively assigns objects to the nearest cluster and updates cluster centroids.
- The k-means method is conceptually intuitive, scalable, and guaranteed to converge to some local optimum, but it requires the manual specification of the number of clusters, and its results can be influenced by the initial means.
- The k-means method may struggle with finding clusters of different sizes and densities and is less effective in high-dimensional data due to the dominance of noise in the Euclidean distance measure.

Variations of k- means

k-Medoids: a
representative object-
based technique

k-Modes: clustering
nominal data

k-Medoids: a representative object-based technique

- The k-means algorithm is sensitive to outliers, and their presence can distort the mean values and impact cluster assignments.
- The k-medoids method addresses the sensitivity to outliers by selecting actual objects as representatives for clusters, reducing the impact of outliers.
- In k-medoids, the sum of dissimilarities between each object and its corresponding representative object is minimized, using an absolute-error criterion.

k-Medoids: a representative object-based technique

- The k-medoid problem is NP-hard when k is a general positive number.
- The Partitioning Around Medoids (PAM) algorithm is a popular realization of k-medoids clustering that iteratively improves cluster quality by replacing representatives and reassigning objects based on the cost function.
- K-medoids is more robust than k-means in the presence of noise and outliers due to the choice of medoids as representatives. However, it is computationally more costly for large values of n and k . Both methods require specifying the number of clusters, k .

k-Modes: clustering nominal data

- The k-means method is limited in cases where the mean of a set of objects is not defined, such as with data containing nominal attributes.
- The k-modes method is a variant of k-means designed for clustering nominal data, where means are replaced with modes.
- The mode in a set of data is the most frequently occurring value.
- In k-modes, distances between objects are computed using a new distance function based on the number of different values on each dimension.

k-Modes: clustering nominal data

- The k-modes method follows a similar iterative process as k-means, involving selecting initial modes, allocating objects to clusters, and updating cluster modes.
- The k-means and k-modes methods can be integrated into the k-prototype method, allowing clustering of data with mixed numeric and nominal values. This method uses different distance measures for numeric and nominal attributes and may involve dimension weights to balance their effects.

Improving Partitioning Methods in Cluster Analysis

- Careful selection of initial cluster centers can enhance the k-means algorithm's convergence and final clustering quality. The k-means++ algorithm, for example, uses a probabilistic approach to choose initial centers.
- Estimating the number of clusters (k) is challenging, but methods like the Calinski-Harabasz index and the gap statistic offer ways to determine an appropriate value for k based on the data's distribution.
- Conventional k-means and its Euclidean distance-based approach are limited to detecting convex, linearly separable clusters. However, kernel k-means overcomes this limitation by using kernel functions like the Gaussian radial basis function to transform similarity measurements among data points, enabling the detection of non-linear, non-convex clusters.

Hierarchical methods

- Partitioning methods organize objects into exclusive groups, while hierarchical clustering creates a hierarchy or "tree" of clusters.
- The section covers hierarchical clustering methods and includes basic concepts, agglomerative (bottom-up) and divisive (top-down) approaches, combining hierarchical clustering with other methods (BIRCH for numeric data), and probabilistic hierarchical clustering methods.

Basic concepts of hierarchical clustering

- Hierarchical clustering organizes data into a hierarchy or "tree" structure, useful for data summarization and visualization.
- Examples include organizing employees in a company into groups or grouping handwritten characters.
- The hierarchy is not assumed to represent the data's inherent structure but is used for summarization and visualization.
- Hierarchical clustering methods can be agglomerative (bottom-up) or divisive (top-down).
- Agglomerative methods merge clusters iteratively, starting with each object as a separate cluster, while divisive methods start with all objects in one cluster and recursively split them.

Basic concepts of hierarchical clustering

- Users can specify the desired number of clusters as a termination condition in both agglomerative and divisive methods.
- Integrating hierarchical clustering with other techniques can improve clustering quality, resulting in multiple-phase clustering.
- Hierarchical clustering methods can be categorized as deterministic (agglomerative, divisive, and multiphase) or probabilistic, using probabilistic models to capture clusters and measure cluster quality.

Agglomerative hierarchical clustering

- Agglomerative hierarchical clustering requires measuring the similarity between clusters.
- Four common linkage measures for distance between clusters are: minimum distance, maximum distance, mean distance, and average distance.
- Minimum distance measure is also called a nearest-neighbor or single-linkage method, sensitive to outliers.
- Maximum distance measure is also called a farthest-neighbor or complete-linkage method, which connects all nodes in clusters.

Agglomerative hierarchical clustering

- Mean and average distance measures offer a compromise between the minimum and maximum distances and can handle both numeric and categorical data.
- Ward's criterion suggests merging clusters to minimize the sum of squared errors (SSE).
- The Lance-Williams algorithm generalizes various proximity measures for clusters, providing a uniform way to specify the distance between merged clusters and others.

Agglomerative hierarchical clustering

- Different parameters in the Lance-Williams formula determine the hierarchical clustering algorithm, e.g., single-linkage, complete-linkage, or Ward's criterion.
- Lance-Williams formula combines similarities between clusters C_i and C_j to cluster C_k through four terms with specific parameters.
- By using the Lance-Williams formula, hierarchical clustering can be generalized for different distance measures and algorithms.

Divisive hierarchical clustering

- Divisive hierarchical clustering partitions objects step by step into clusters.
- Three important considerations when designing a divisive hierarchical clustering method:
 1. Choice of splitting criterion: Determines which splitting is the best. Different criteria include SSE for numeric data and the Gini index for nominal data.
 2. Designing a splitting method: Balancing optimization and computational cost, e.g., using heuristic or approximate methods.
 3. Selecting which cluster to split next: Intuitively, choosing the "loosest" cluster based on average SSE.

Divisive hierarchical clustering

- The minimum spanning tree-based approach involves constructing a weighted graph where each point is a node, and the distance between two points is the weight of the edge connecting them.
- The minimum spanning tree represents the most compact way to connect points into one cluster, and its subtrees represent individual clusters in the divisive hierarchical clustering process.
- The splitting criterion in this method is the total weights of all edges in the spanning trees of clusters, with smaller values being preferred.

Divisive hierarchical clustering

- The splitting method involves deleting the edge with the largest weight in the minimum spanning tree to divide a cluster.
- A dendrogram is a tree structure used to represent the hierarchical clustering process. It shows how objects are grouped or partitioned step by step.
- Dendrograms can be used to visualize and analyze hierarchical clustering results, but small or numerous clusters may be ignored in practice.

BIRCH: scalable hierarchical clustering using clustering feature trees

- Agglomerative and divisive hierarchical clustering methods have limitations, including the inability to revisit merge or split decisions and scalability issues.
- Hierarchical clustering can be conducted in multiple phases to improve clustering results. BIRCH is a method designed to address these issues, especially for clustering large amounts of numeric data.
- BIRCH uses clustering features and clustering feature trees (CF-trees) to represent clusters in an efficient and scalable way.

BIRCH: scalable hierarchical clustering using clustering feature trees

- A clustering feature (CF) summarizes cluster statistics, including the linear sum (LS) and square sum (SS) of data points.
- CFs enable the computation of cluster properties like centroid, radius, and diameter.
- Clustering features avoid the need to store detailed information about individual objects, contributing to the efficiency of BIRCH.
- CFs are additive, which allows for efficient merging of clusters.

BIRCH: scalable hierarchical clustering using clustering feature trees

- A CF-tree is a height-balanced tree structure storing clustering features for hierarchical clustering. It has parameters like branching factor and threshold to control its size.
- BIRCH minimizes input/output (I/O) time and applies multiphase clustering, including Phase 1 (building an initial CF-tree) and Phase 2 (clustering leaf nodes of the CF-tree).
- Phase 1 is incremental, and the CF-tree is built dynamically as objects are inserted. Objects are inserted into the closest leaf entry, and nodes are split if the diameter exceeds the threshold.

BIRCH: scalable hierarchical clustering using clustering feature trees

- Rebuilding the CF-tree is performed when needed, and this process does not require rereading all the data.
- BIRCH demonstrates linear scalability with respect to the number of objects and good clustering quality in experiments.
- However, BIRCH may not perform well for non-spherical clusters due to its reliance on the radius or diameter to control cluster boundaries.
- The concepts of clustering features and CF-trees have been applied to address clustering streaming and dynamic data problems beyond BIRCH.

Probabilistic hierarchical clustering

- Hierarchical clustering methods using linkage measures are commonly used and efficient but have drawbacks.
- Drawbacks include the challenge of choosing a suitable distance measure, inability to handle missing attribute values in data, and the heuristic nature of decision-making during merging and splitting steps.
- Probabilistic hierarchical clustering seeks to address these drawbacks by using probabilistic models to measure distances between clusters.

Probabilistic hierarchical clustering

- Clustering is viewed as an estimation problem where data objects are seen as a sample from an underlying generative model.
- Probabilistic models, such as Gaussian distributions, are used to estimate generative models from the observed data.
- Clusters' quality is measured using maximum likelihood and represented by the maximum likelihood probability ($P()$).
- Clustering quality can be measured by the change in quality when merging or splitting clusters.

Probabilistic hierarchical clustering

- A probabilistic hierarchical clustering method can use a probabilistic framework to measure cluster similarity.
- Merging two clusters is considered based on maximizing log likelihood.
- The method starts with one cluster per object and iteratively merges clusters that lead to improvements in clustering quality.
- Probabilistic hierarchical clustering methods are easy to understand and have efficiency comparable to agglomerative hierarchical clustering.

Probabilistic hierarchical clustering

- They can handle partially observed data, making them suitable for datasets with missing attribute values.
- However, they output a single hierarchy with respect to a chosen probabilistic model and do not handle the uncertainty of cluster hierarchies.
- Advanced probabilistic clustering methods like Bayesian tree-structured models are used for handling such complexities.

Density-based and grid-based methods

- Most partitioning and hierarchical clustering methods are designed for finding spherical-shaped clusters and face difficulties with clusters of arbitrary shapes.
- Feature transformation methods, like kernel k-means, may help in finding non-spherical clusters, but choosing the right kernel functions can be challenging.
- For data with arbitrary-shaped clusters, density-based clustering methods are an alternative approach, focusing on modeling clusters as dense regions separated by sparse regions.
- Two representative density-based clustering methods are DBSCAN and DENCLUE
- To address the computational cost of density-based clustering, the data space can be partitioned into a grid, leading to the development of grid-based clustering methods

DBSCAN: density-based clustering based on connected regions with high density

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise) finds dense regions in density-based clustering by considering core objects, which are objects with dense neighborhoods, and their connections to form clusters.
- DBSCAN uses two user-specified parameters, ϵ and MinPts, where ϵ specifies the neighborhood radius, and MinPts determines the density threshold for core objects.
- Core objects are identified based on the condition that their ϵ -neighborhood contains at least MinPts objects. Non-core objects are labeled as noise points.

DBSCAN: density-based clustering based on connected regions with high density

- Clusters are formed by connecting core objects and their respective neighborhoods, ensuring that the clusters consist of density-connected core objects.
- The clustering process starts by selecting an unvisited object, marking it as visited, and checking its ϵ -neighborhood. Objects are iteratively added to the cluster until it can no longer be expanded, at which point it is considered a completed cluster.
- The algorithm continues until all objects are visited, and it finds clusters in arbitrary shapes.

DBSCAN: density-based clustering based on connected regions with high density

- The computational complexity of DBSCAN depends on whether a spatial index is used: $O(n \log n)$ with an index or $O(n^2)$ without.
- DBSCAN can be extended to hierarchical density-based clustering using HDBSCAN, and it achieves density-based hierarchical clustering by taking only one parameter, MinPts.
- In HDBSCAN, the mutual reachability distance between two objects is used to construct a mutual reachability graph, which can be analyzed to find density-based hierarchical clusters.

DBSCAN: density-based clustering based on connected regions with high density

- OPTICS is another density-based clustering method that provides a cluster ordering, representing the density-based clustering structure of the data. It doesn't require the user to specify a density threshold and can be used to extract clustering information, derive intrinsic structures, and provide visualizations of clusters.
- OPTICS processes objects in an order that selects objects with lower ϵ values to ensure clusters with higher density are finished first. It generally produces a hierarchical clustering structure instead of simple data partitioning like DBSCAN.

DENCLUE: clustering based on density distribution functions

- DENCLUE, which stands for DENsity-based CLUs- tEring, is a density-based clustering method that uses kernel density estimation to identify density attractors, representing regions of high-probability density in the data.
- Kernel density estimation is a statistical approach employed in DENCLUE to estimate probability density. It quantifies the influence of nearby data points within a specified bandwidth, allowing for the creation of density functions.
- DENCLUE uses a Gaussian kernel function to estimate density based on the input dataset.

DENCLUE: clustering based on density distribution functions

- Density attractors in DENCLUE are points considered to be local maxima of the estimated density function. These density attractors serve as cluster centers.
- The method utilizes a noise threshold (ξ) to distinguish between nontrivial density attractors (cluster centers) and noise. Only density attractors with $\hat{f}(x^*) \geq \xi$ are considered as nontrivial density attractors.
- Objects are assigned to clusters through a stepwise hill-climbing procedure guided by the gradient of the estimated density function. The procedure stops when the gradient decreases, and an object is assigned to a specific density attractor. Outliers or noise points are identified during this process.

DENCLUE: clustering based on density distribution functions

- In DENCLUE, a cluster is defined as a combination of density attractors (X) and the objects assigned to them (C). To be part of the same cluster, every object in C must have a path connecting them to density attractors in X with density values above ξ .
- DENCLUE has the advantage of being invariant against noise. The kernel density estimation method employed effectively reduces the impact of noise by uniformly distributing it within the input data.

Grid-based methods

- Grid-based clustering methods offer a solution to the efficiency and scalability challenges often encountered in clustering, particularly with large and high-dimensional datasets.
- These methods employ a multiresolution grid data structure that divides the object space into cells, providing a framework for clustering operations. The processing time of grid-based clustering is determined by the number of cells, rather than the number of data objects.
- Grid-based clustering typically involves three main steps:
 1. Creation of a grid structure to partition the data space into cells.
 2. Calculation of cell densities for each cell, allowing for efficient processing.
 3. Assembly of clusters using the dense cells, which can be of arbitrary shape.

Grid-based methods

- CLIQUE is an example of a grid-based method designed for identifying density-based clusters in subspaces. It partitions each dimension into non-overlapping intervals, identifies dense cells based on a density threshold, and utilizes a monotonicity property with respect to dimensionality to guide the search for dense cells.
- CLIQUE's effectiveness lies in its ability to automatically discover high-density clusters in subspaces of various dimensionalities, offering scalability and sensitivity to data distribution.

Grid-based methods

- However, proper tuning of grid size and density threshold is crucial for meaningful results, and there may be overlaps among reported dense regions, limiting its capability to detect clusters of different densities within different dimensional subspaces.
- STING is another grid-based multiresolution clustering technique that uses a hierarchical structure of rectangular cells. It precomputes and stores statistical parameters for query processing, which are useful for identifying density-based clusters when granularity approaches zero.

Grid-based methods

- STING has several advantages, including query-independence, parallel processing, incremental updating, and high efficiency. It can approximate the clustering result of DBSCAN as granularity approaches zero.
- However, the quality of STING clustering depends on the granularity, and the technique tends to produce clusters with isothetic boundaries, reducing the quality and accuracy of results despite its efficiency.

Evaluation of clustering

- Assessing clustering tendency: Determines if a dataset exhibits nonrandom structure, making clustering analysis meaningful.
- Determining cluster count: Estimating the optimal number of clusters, a critical dataset summary and parameter for algorithms like k-means.
- Measuring clustering quality: Evaluates the quality of resulting clusters through various measures, assessing their fit, alignment with ground truth (if available), and enabling comparisons between different clustering results on the same dataset.

Assessing clustering tendency

- Clustering tendency assessment distinguishes random from meaningful data structures that lead to useful clusters.
- Determining clustering tendency involves measuring the probability of data generated by a uniform distribution.
- The Hopkins Statistic assesses spatial randomness by comparing real data with uniformly distributed points.

Assessing clustering tendency

- To calculate the Hopkins Statistic:
 - Sample n points from the dataset, finding the distance (x_i) between each point and its nearest neighbor.
 - Sample n points uniformly from the dataset, finding the distance (y_i) between each point and its nearest neighbor.
 - Calculate H using the formula $H = \sum(x_i) / (\sum(x_i) + \sum(y_i))$, with ' d ' being the dimensionality of the dataset.

Assessing clustering tendency

- The Hopkins Statistic (H) helps determine whether data follows a uniform distribution:
 - $H \approx 0.5$ suggests uniform distribution.
 - $H \approx 1$ indicates significant skewness, implying meaningful clustering.
- Other methods like spatial histogram and distance distribution also compare statistics between data and random distributions for clustering tendency analysis.

Determining the number of clusters

- Determining the optimal number of clusters is vital to strike a balance between data compression and analysis accuracy.
- Selecting the correct number of clusters depends on data distribution and user requirements.
- A straightforward method is to set the number of clusters to $\sqrt{n/2}$ for n data points.
- The Calinski-Harabasz index, discussed in Section 8.2.2, helps estimate the number of clusters for k-means.

Determining the number of clusters

- The elbow method identifies the turning point in the curve of within-cluster variances concerning the number of clusters.
- Advanced techniques use information criteria or information theoretic approaches for cluster number estimation.
- Cross-validation can determine the appropriate cluster count by dividing the data into subsets, building clustering models on subsets, and evaluating model quality with a test set.
- Repeating this process for different k values allows comparison and the selection of the optimal number of clusters.

Measuring clustering quality: extrinsic methods

- Extrinsic methods evaluate clustering quality by comparing results to ground truth (ideal clustering).
- Extrinsic methods can be categorized as either matching-based (supervised) or information theory-based (supervised) if ground truth is available.
- Desiderata for effective measures include cluster homogeneity, cluster completeness, rag bag handling, and small cluster preservation.
- Matching-based methods, such as purity, focus on matching clusters with ground truth groups.

Measuring clustering quality: extrinsic methods

- Information theory-based methods use conditional entropy to compare clustering results to ground truth.
- Pairwise comparison-based methods assess quality by comparing object pairs' assignments to clusters and ground truth.
- Various measures like Jaccard coefficient, Rand statistic, Fowlkes-Mallows measure, BCubed precision, and recall are employed for pairwise comparison-based evaluation.
- Correlation analysis can also be used to measure the quality of clustering results by examining element-wise correlations between binary matrices derived from the ground truth and clustering results.

Intrinsic methods

- Intrinsic methods assess clustering quality without external ground truth and focus on cluster compactness and separation.
- The Dunn index measures compactness using the maximum distance between two points in the same cluster (δ) and separation using the minimum distance between points in different clusters (γ). The index is calculated as $DI = \gamma / \delta$.
- The silhouette coefficient, another intrinsic measure, considers the average distance between objects in their cluster ($a(o)$) and the minimum average distance to other clusters ($b(o)$) for each object o .

Intrinsic methods

- The silhouette coefficient ranges from -1 to 1, where positive values indicate well-separated and compact clusters, while negative values suggest poor clustering.
- The average silhouette coefficient of all objects in a cluster measures the cluster's fitness. The quality of a clustering can be assessed using the average silhouette coefficient for all objects in the dataset.
- These intrinsic measures, including the silhouette coefficient, can also be employed in the elbow method to estimate the number of clusters in a dataset.

Summary

- Clusters group similar data objects together while keeping them dissimilar from objects in other clusters, forming the basis of cluster analysis.
- Cluster analysis has diverse applications in fields like business intelligence, biology, and image recognition, serving as both a standalone tool and a preprocessing step for other data mining algorithms.
- It's a challenging field, requiring scalability, the handling of various data types, dealing with noisy data, and more, often without requiring substantial domain knowledge.

Summary

- Several clustering algorithms exist, categorized by factors such as partitioning criteria, similarity measures, and clustering space.
- Major clustering methods include partitioning methods (e.g., k-means), hierarchical methods (agglomerative and divisive), and density-based and grid-based methods.
- Partitioning methods create k initial clusters and iteratively refine them, including centroid-based (k-means), object-based (k-medoids), and mode-based (k-modes) techniques.

Summary

- Hierarchical methods create a hierarchical decomposition of data objects, employing linkage measures such as single-linkage and complete-linkage. BIRCH combines hierarchical and other methods.
- Probabilistic hierarchical clustering utilizes probabilistic models to measure distances between clusters.
- Density-based methods cluster objects based on density, growing clusters according to neighborhood or density functions. OPTICS is an example.

Summary

- Grid-based methods quantize the object space into a grid structure and then perform clustering. Examples include STING and CLIQUE.
- Clustering evaluation is essential, involving assessing clustering tendency, determining the number of clusters, and measuring clustering quality.
- Evaluation methods include statistics like the Hopkins statistic, Calinski-Harabasz index, elbow method, and intrinsic measures like the Dunn index and silhouette coefficient. Extrinsic and intrinsic methods exist depending on the availability of ground truth.